

Acute ankle injuries

with PIETER D'HOOGE, EVERT VERHAGEN and JON KARLSSON

Total rupture of left ATFL (ankle ligament) and associated joint capsule damage in a soccer kickabout with friends. Continuing to assess extent of injury and treatment plan day by day. Rehab already started.

World golf number 1 Rory McIlroy on Instagram, nine days before the British Open, 6 July 2015

Ankle injury is arguably the most common sporting injury. In 24 of the 70 sports for which there are quality data, ankle injury holds the number 1 spot.¹ In sports such as volleyball, ankle injuries account for nearly half of all injuries.²

Ankle injuries include, but are not limited to 'ankle sprains'. The first half of this chapter focuses on anatomy, clinical assessment and management of lateral ligament injuries after ankle sprain and their sequelae. We address less common immediate diagnoses for ankle sprains: medial ligament injury and significant ankle fractures.

Importantly for clinicians who work in sport, a 'sprained ankle' can mask damage to other structures, such as subtle fractures around the ankle joint, osteochondral fractures of the dome of the talus, and dislocation or longitudinal rupture of the peroneal tendons. Such injuries and their complaints persist longer than would be expected with a straightforward lateral ligament sprain. This is often referred to as 'the difficult ankle' and it is the focus of the second half of this chapter.

FUNCTIONAL ANATOMY

The ankle contains three joints (Fig. 41.1):

1. talocrural (ankle) joint
2. inferior tibiofibular joint (syndesmosis)
3. subtalar joint.

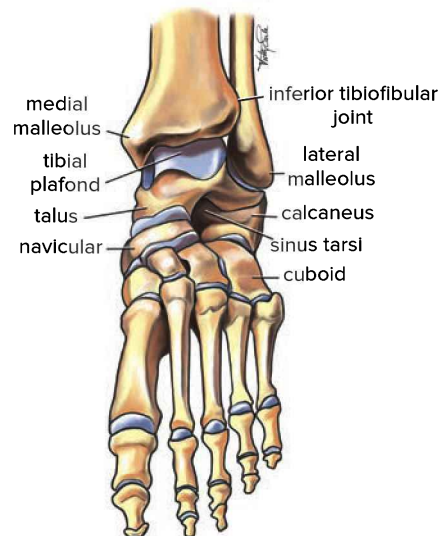
The talocrural or ankle joint (Fig. 41.1a) is a hinge joint formed between the inferior surface of the tibia and the superior surface of the talus.

The medial and lateral malleoli provide additional articulations and stability to the ankle joint. The ankle joint can plantarflex and dorsiflex. As the ankle is least stable in plantarflexion, where most stability is provided

by the ligaments, injuries are more common when the foot is in this position.

The distal parts of the fibula and tibia articulate at the inferior tibiofibular joint where they are supported by the inferior tibiofibular ligaments or syndesmosis. The small amount of movement present at this joint is extremely important for normal walking and running. Injuries to this joint are more common than previously recognised.

The subtalar joint (Fig. 41.1b), between the talus and calcaneus, is divided into an anterior and posterior articulation separated by the sinus tarsi.



(a)

Figure 41.1 Anatomy of the ankle. (a) Talocrural (ankle) joint

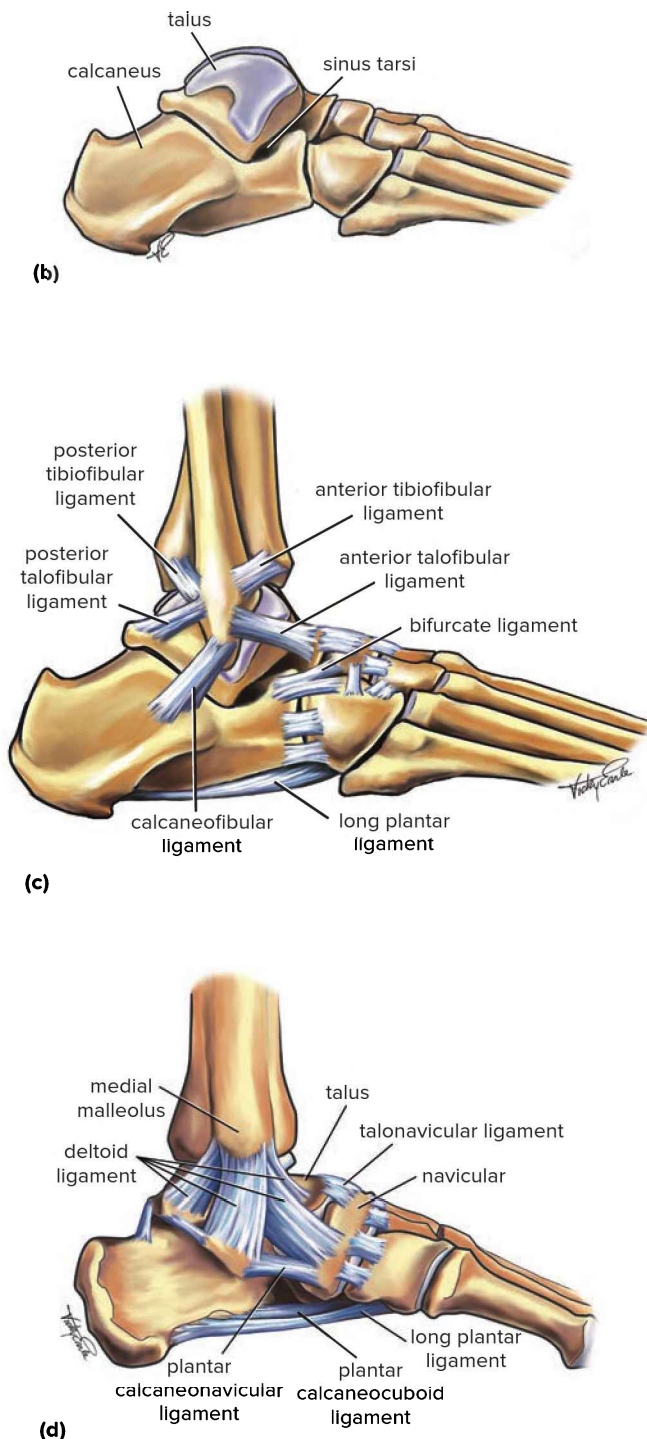


Figure 41.1 (cont.) (b) Subtalar joint (c) Ligaments of the ankle-lateral view (d) Ligaments of the ankle-medial view

Inversion and eversion occur at the subtalar joint. It provides shock absorption and permits the foot to adjust to uneven ground. The subtalar joint allows the foot to remain flat on the ground when the leg is at an angle to the surface.

The ligaments of the ankle joint are shown in Figures 41.1c and d.

The lateral ligament consists of three parts:

- anterior talofibular ligament (ATFL), which passes as a flat and rather thin band from the anterior aspect of the fibula anteriorly to the lateral talar neck
- calcaneofibular ligament (CFL), a cord-like structure directed inferiorly and posteriorly
- posterior talofibular ligament (PTFL), a short structure running posteriorly from the fibula to the talus.

The medial or deltoid ligament of the ankle is a fan-shaped ligament extending from the medial malleolus anteriorly to the navicular and talus, inferiorly to the calcaneus and posteriorly to the talus.

It can be differentiated into the anterior tibiotalar ligament (most commonly affected), tibiocalcaneal ligament, posterior tibiotalar ligament and the tibionavicular ligament. This strong ligament is composed of two layers, one deep and the other more superficial.

The syndesmosis consists of the anterior and posterior inferior tibiofibular ligament (AITFL/PITFL) and the interosseous tibiofibular ligament (IOTFL). In almost 75% of cases, there are contact facets with articular cartilage between the distal tibia and fibula, forming a true synovial joint. The PITFL provides approximately 40% of the resistance to lateral displacement, the AITFL 35% and the IOTFL 22%. The remaining stability is provided by the interosseous membrane.

Relevant nerves for the ankle originate from the lower lumbar and higher sacral spinal roots (L4-S2). Efferent fibres of the deep peroneal nerve innervate the anterior muscles of the leg, the superficial peroneal nerve innervates the peroneal muscles on the lateral side, and the tibial nerve innervates the muscles on the posterior side of the leg. The afferent path consists of all nerves that send proprioceptive information from mechanoreceptors around the ankle to the central nervous system.

CLINICAL PERSPECTIVE

It is estimated that 10 ankle sprains occur per 1000 exposures.¹ Inversion injuries are at least four times more common than eversion injuries due to the relative instability of the lateral joint and weakness of the lateral ligaments compared with the medial ligament.³ Around 80% of ankle sprains are ligamentous caused by sudden inversion or supination.¹ The injury motion often happens at the subtalar joint and tears the ATFL which is the weakest of the lateral ankle ligaments.

Table 41.1 Acute ankle injuries

Common	Less common	Not to be missed
Lateral ligament sprain • ATFL • CFL • PTFL	Osteochondral lesion of the talus Ligament sprain/rupture • Medial ligament injury • AITFL injury Fractures • Lateral/medial/posterior malleolus (Pott's—bimalleolar) • Tibial plafond • Base of the fifth metatarsal • Anterior process of the calcaneus • Lateral process of the talus • Posterior process of the talus • Os trigonum Dislocated ankle (fracture/dislocation) Tendon rupture/dislocation • Tibialis posterior tendon • Peroneal tendons (longitudinal rupture)	Complex regional pain syndrome type 1 (post-injury) Greenstick fractures (children) Syndesmosis injury Tarsal coalition (may come to light as a result of an ankle sprain)

ATFL = anterior talofibular ligament; AITFL = anteroinferior tibiofibular ligament; CFL = calcaneofibular ligament; PTFL = posterior tibiofibular ligament

As the stronger medial ligament requires a greater force to be injured, these sprains almost always take longer to rehabilitate. The differential diagnoses that must be considered after an ankle injury are listed in Table 41.1. The aim of the initial clinical assessment is to rule out ankle fractures (see Ottawa ankle rules, below), and if possible, to diagnose the site of abnormality as accurately as possible.

History

PRACTICE PEARL

The mechanism of the ankle sprain is an important clue to diagnosis. An inversion injury suggests lateral ligament and medial compression damage. Eversion injury suggests medial ligament damage. If the injury involved ankle compression, the clinician should consider osteochondral injury.

The onset of pain is very important. A history of initial ability to weight-bear after an injury followed by a subsequent increase in pain and swelling with continued activity suggests a ligament injury rather than a fracture. The location of pain and swelling generally indicates which ligaments were injured. The most common site is over the anterolateral aspect of the ankle involving the ATFL, in approximately two-thirds of all injuries.

In inversion injuries the ATFL is the first ligament to tear, and is torn in 97% of cases. The CFL can then subsequently

tear after the ATFL. Isolated rupture of the CFL occurs in only 3% of inversion injuries. The last ligament to tear is the PTFL, which is usually uninjured unless there is a frank dislocation of the ankle. In severe injuries, both medial and lateral ligaments may be damaged; however, this is infrequent. The degree of swelling and bruising is usually, but not always, an indication of the severity of injury.

The degree of disability, both immediately following the injury and subsequently, indicates the severity of the injury (Fig. 41.2). The practitioner should ask about initial management, the use of the POLICE regimen (Chapter 17) and the duration of restricted weight-bearing after the injury, as a poorly managed injury will appear 'more significant' in terms of swelling and long-term disability than a similar injury that has been managed appropriately. An example of severe swelling and bruising in an ankle managed without POLICE is shown in Figure 41.2.

The practitioner should ask about a previous history of ankle injury and assess whether the post-injury rehabilitation was adequate. Did the athlete use protective tape or braces for the first 3–6 months after the previous injury, or until the end of the season?

Examination

The aims of ankle examination are to:

- assess the degree of instability present (grade the ligamentous injury)
- detect functional deficits (i.e. loss of range of motion, reduced strength and reduced proprioception)



Figure 41.2 Severe swelling and bruising in an ankle

- detect any associated injuries (e.g. avulsion fracture of the base of the fifth metatarsal is commonly overlooked but is easily detected by palpation. Also, injury to the peroneal tendons may lead to longstanding disability and is easily missed during the acute phase.).

Examination involves:

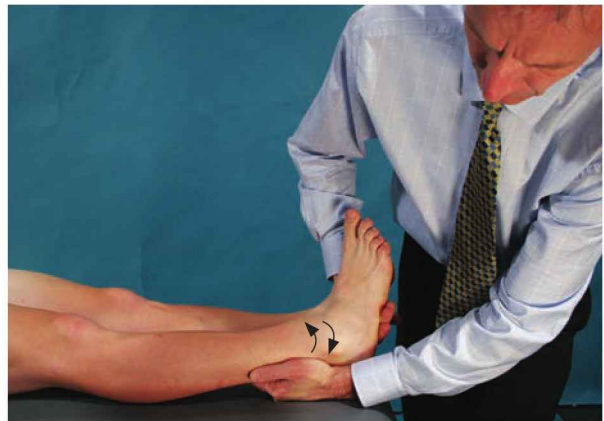
1. Observation
 - a. standing
 - b. supine
2. Active movements
 - a. plantarflexion/dorsiflexion (Fig. 41.3a)
 - b. inversion/eversion



(a)

Figure 41.3 (a) Active movement—plantarflexion/dorsiflexion. Assessment of dorsiflexion is important as restriction results in a functional deficit. Range of motion can be compared with the uninjured side. Tight calf muscles may restrict dorsiflexion. This can be eliminated by placing the knee in 70–90° flexion

3. Passive movements
 - a. plantarflexion/dorsiflexion
 - b. inversion/eversion (Fig. 41.3b)
4. Resisted movements
 - a. eversion (Fig. 41.3c)
5. Functional tests
 - a. lunge test (Fig. 41.3d)
 - b. hopping
6. Palpation
 - a. distal—and proximal—fibula
 - b. lateral malleolus
 - c. lateral ligaments (Fig. 41.3e)

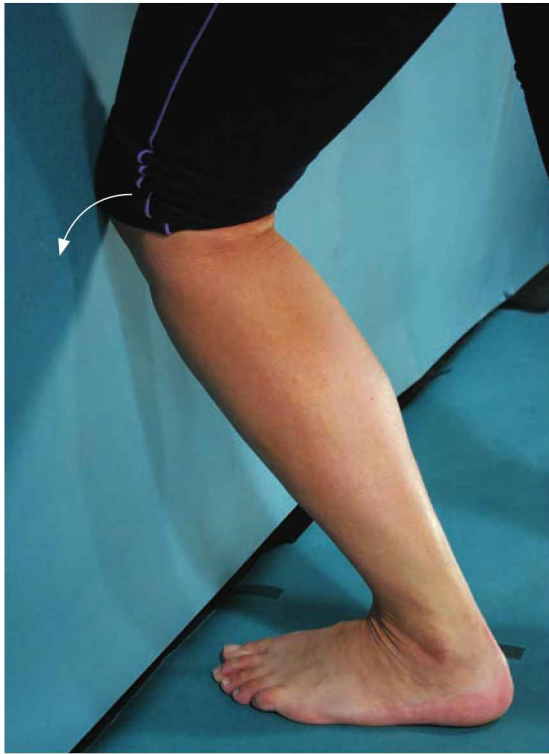


(b)



(c)

(b) Passive movement—inversion/eversion. Inversion is frequently painful and restricted in lateral ligament injury, while eversion is painful following injuries to the medial ligament. Increased pain on combined plantar flexion and inversion suggests ATFL injury (c) Resisted eversion—this can be difficult in the acute setting. In cases of persistent pain, weakness of the evertors (peroneal muscles) should be assessed



(d)



(e)

Figure 41.3 (cont.) (d) Functional test—lunge test. Assess ankle dorsiflexion compared with the uninjured side. Note any pain. Other functional tests may be performed to reproduce the patient's pain if appropriate (e.g. single-leg standing, hopping) (e) Palpation—lateral ligament

- d. talus
- e. peroneal tendon(s)
- f. base of fifth metatarsal
- g. anterior joint line

- h. medial ligament
- i. sustentaculum tali
- j. sinus tarsi
- k. anteroinferior tibiofibular ligament

7. Special tests (comparison with other side necessary; however, it should be borne in mind that there is no obvious correlation between increased anterior drawer and/or lateral talar tilt and symptoms of ankle insufficiency)
 - a. anterior drawer (Fig. 41.3f)
 - b. lateral talar tilt (increased inversion) (Fig.41.3g)
 - c. proprioception (Fig. 41.3h).

Investigations

Should the clinician order a radiograph? Clinicians can use Ottawa ankle rules to assist them in making this decision (Fig. 41.4).^{4, 5}

Radiographs of the ankle joint should include the base of the fifth metatarsal to exclude avulsion fractures. If damage to the lower tibiofibular syndesmosis (AITFL) is suspected, special ankle mortise or syndesmosis views are required.

An osteochondral lesion of the talus—especially the medial side—may not be apparent on initial radiograph. If significant pain and disability persist despite appropriate



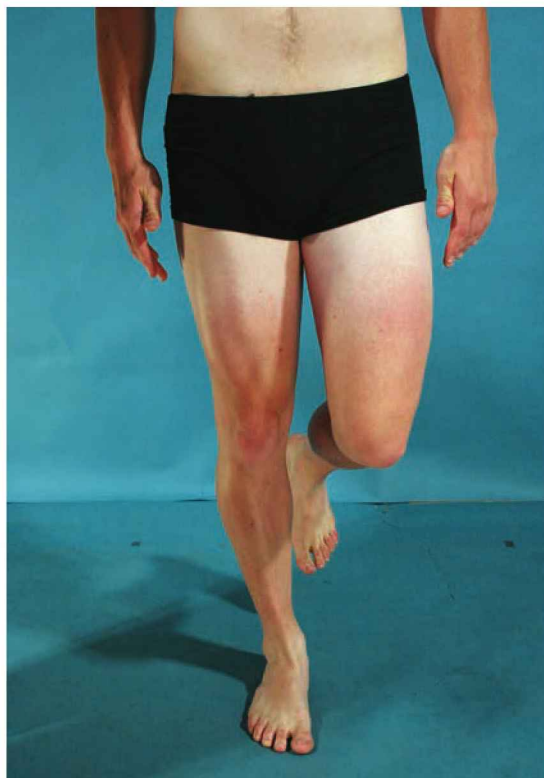
(f)

(f) Ligament testing—anterior drawer test. The ankle is placed in slight plantarflexion and grasped as shown. Pressure is exerted upwards and the degree of excursion (anterior drawer) is noted and compared with the uninjured side. This test assesses the integrity of the ATFL and CFL. Pain on testing should also be noted; if painful it may indeed mask injury to the ligament. Then the test should be repeated after five days. The optimal time to test the integrity of the lateral ligaments is on the fifth post-injury day



(g)

Figure 41.3 (cont.) (g) Ligament testing—talar tilt. This tests integrity of the anterior talofibular and calcaneofibular ligaments laterally and the deltoid ligament medially. The ankle is grasped as shown and the medial and lateral movement of the talus and calcaneus are assessed in relation to the tibia and fibula. Pain on this test must also be noted (h) Special test—proprioception. Single-leg standing with eyes closed may demonstrate impaired proprioception compared with the uninjured side



(h)

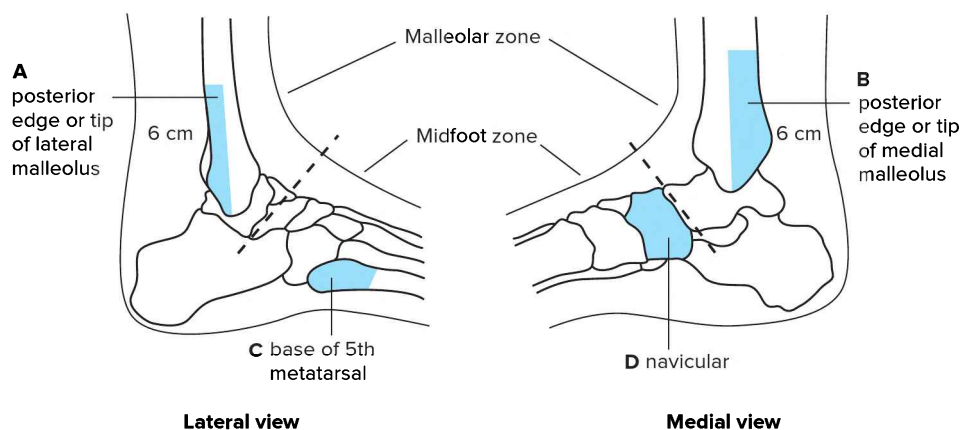


Figure 41.4 Ottawa ankle rules X-ray recommendation⁴—Ankle X-rays are only required if there is any pain in the malleolar zone, and any one of these findings: bone tenderness at A; bone tenderness at B; inability to bear weight both immediately and at the clinical assessment (four steps). Foot X-ray recommendation—foot X-rays are only required if there is any pain in the midfoot zone, and any one of these findings: bone tenderness at C; bone tenderness at D; inability to bear weight both immediately and at the clinical assessment (four steps)

treatment 4–6 weeks after an apparent 'routine' ankle sprain, MRI is the investigation of choice to exclude an osteochondral lesion, especially when dealing with

the 'difficult ankle', that is, longstanding pain/disability without obvious cause or pathology. If MRI is not available a radioisotopic bone scan or CT can be used.

LATERAL LIGAMENT INJURIES

Lateral ligament injuries often occur during activities requiring rapid changes in direction, especially if these take place on uneven surfaces (e.g. grass fields). They also occur when a player, having jumped, lands on another competitor's foot. They are very common in basketball, volleyball, netball and most football codes.

The usual mechanism of lateral ligament injury is inversion and plantarflexion, and this damages the ATFL before the CFL.⁶ This occurs because the ATFL is taut in plantarflexion and the CFL is relatively loose (Fig. 41.5).

Also, the ATFL can only tolerate half the strain of the CFL before tearing. A complete tear of the ATFL, CFL and PTFL is rare and results in a dislocation of the ankle joint (Fig. 41.6). Isolated ligament ruptures of the CFL and especially the PTFL are also rare.

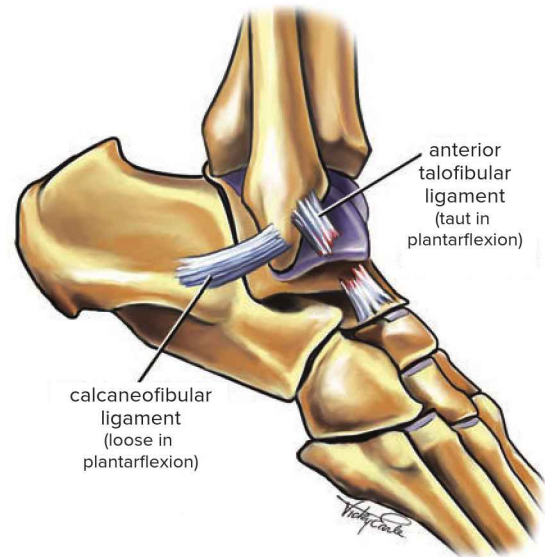


Figure 41.5 A plantarflexion injury generally leads to injury of the anterior talofibular ligament before the calcaneofibular ligament



Figure 41.6 A posteromedial ankle dislocation

Ankle sprain may be accompanied by an audible snap, crack or tear, which, although often of great concern to the athlete and onlookers, has no particular diagnostic significance. This is unlike the case in knee ligament injuries where a 'pop' has profound implications (see Chapter 35). Depending on the severity of the injury, the athlete may continue to play or may have to stop immediately. Swelling usually appears rapidly, although occasionally it may be delayed some hours.

To assess lateral ligament injuries, examine all three components of the ligament and determine the degree of ankle laxity:

- A grade I tear has normal ligament laxity. It is important to compare both sides (assuming the other side has not been previously injured) as there is a large inter-individual variation in normal ankle laxity.
- A grade II tear reveals some degree of laxity but with a firm end point.
- A grade III tear shows gross laxity without a discernible end point.

All three grades are associated with pain and tenderness, although grade III tears may be least painful after the initial episode has settled. Grading of these injuries guides prognosis and helps determine the rate of rehabilitation. Grading also affects selection of the acute treatment. Note that in the acute phase, the reliability and validity of manual stress tests (e.g. anterior drawer or talar-tilt test) are very low. Manual stress tests are more reliable and valid when performed 5-7 days post injury. But keep in mind the limitations of 'special tests' in general (Chapter 14).⁷

TREATMENT AND REHABILITATION OF LATERAL LIGAMENT INJURIES

The management of lateral ligament injuries regardless of the grade follows the same principles. After minimising initial haemorrhage and reducing pain, the aims are to restore range of motion, muscle strength and proprioception, and then prescribe a progressive, sport-specific exercise program.

Initial management

Lateral ligament injuries require POLICE treatment where the letters ICE represent ice, compression and elevation (Chapter 17). This limits the haemorrhage and subsequent oedema that can cause an irritating synovial reaction and restrict joint range of motion for a long period of time. The injured athlete should avoid things that promote blood flow and swelling, such as hot showers, heat rubs or excessive weight-bearing. Gradually increased weight-bearing will, however, help

reduce the swelling and increase the ankle motion, and enhances recovery.

Reduction of pain and swelling

Analgesics and/or NSAIDs may be required. After 48 hours, gentle soft tissue therapy and mobilisation may reduce pain. By reducing pain and swelling, muscle inhibition around the joint is minimised, permitting the patient to begin range of motion exercises.

The indications for the use of NSAIDs in ankle injuries are unclear. Many clinicians prescribe these drugs after lateral ligament sprains although their efficacy has not been proven (Chapter 17). The rationale for commencing NSAIDs 2-3 days after injury is to reduce the risk of joint synovitis with early return to weight-bearing.

Restoration of full range of motion

If necessary, the patient may be non weight-bearing on crutches for the first 24 hours, but then should commence partial weight-bearing in normal heel-toe gait. This can be achieved while still using crutches or, in less severe cases, by protecting the damaged joint with strapping or bracing.⁸ Thus, partial and, ultimately, full weight-bearing can take place without aggravating the injury. Lunge stretches, and accessory and physiological mobilisation of the ankle (Fig. 41.7a), subtalar (Fig. 41.7b) and midtarsal joints should begin early in rehabilitation.

As soon as pain allows, the clinician should prescribe active range of motion exercises (e.g. stationary cycling).

Muscle conditioning

Active strengthening exercises, including plantarflexion, dorsiflexion, inversion and eversion (Fig. 41.8), should begin as soon as pain allows.

The exercises should be progressed by increasing resistance (a common method is to use rubber tubing). Strengthening eversion with the ankle fully plantarflexed is particularly important in the prevention of future lateral ligament injuries. Weight-bearing exercises (e.g. shuttle [Fig. 41.9] and wobble-board exercises [Fig. 41.10a]) are encouraged as soon as pain permits, preferably the first or second day after injury.

Proprioception

Proprioception is invariably impaired after ankle ligament injuries. The assessment of proprioception is shown in Figure 41.3h. The patient should begin proprioceptive retraining (Chapter 18) early in rehabilitation and these exercises should gradually progress in difficulty. An example of a common progression is balancing on one leg, then using the wobble board (Fig. 41.10a) or mini-trampoline, and ultimately performing functional activities while balancing (Fig. 41.10b).



(a)



(b)

Figure 41.7 Mobilisation of the ankle joint (a) ankle dorsiflexion. The calcaneus and foot are grasped to passively dorsiflex the ankle (b) eversion mobilisation techniques to restore subtalar joint movement after ankle sprain

Functional exercises

Functional exercises (e.g. jumping, hopping, twisting, figure-of-eight running) can be commenced once an athlete is pain-free, has full range of motion, and adequate muscle strength and proprioception. Specific technical training can accelerate a player's return to sport and substantially reduce the risk of re-injury.^{9, 10, 11} Remember that approximately 75% of those who sustain an ankle ligament injury have had a previous injury. In many cases this old injury was not fully rehabilitated.

PRACTICE PEARL

The high risk of recurrent injury means that all ankle ligament injuries should be fully rehabilitated.



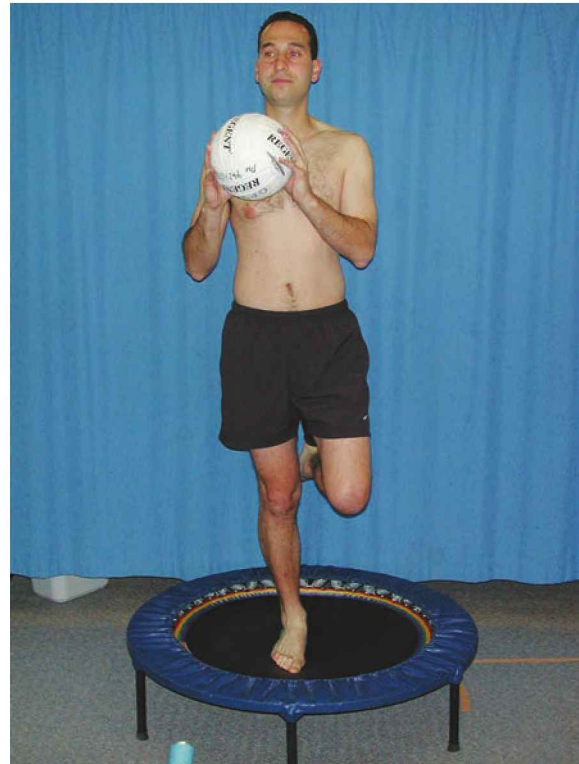
Figure 41.8 Strengthening exercises—eversion using a rubber tube as resistance



Figure 41.9 Strengthening exercises—weight-bearing shuttle exercises



(a)



(b)

Figure 41.10 (a) Proprioceptive retraining following acute ankle injury using wobble board (b) Proprioceptive retraining following acute ankle injury using functional activity while balancing

Return to sport

Return to sport is advised when functional exercises can be performed without pain during or after activity. However, any athlete who has had a significant lateral ligament injury has an increased risk of injury recurrence post injury for a minimum of 6–12 months post-injury.^{2, 12, 13} There are a number of methods to protect against these subsequent inversion injuries. Both external prophylactic measures (i.e. tape or brace), as well as neuromuscular training, seem equally effective in reducing the increased risk of ankle sprain recurrences after an index ankle sprain.^{14, 15} They have seemingly different pathways through which they achieve this secondary preventive effect.

The relative advantages of taping and bracing have been discussed in Chapter 12. As both seem equally effective, the choice of taping or bracing depends on patient preference, cost, availability and expertise in applying tape.¹⁶

The three main methods of tape application are stirrups (Fig. 41.11a), heel lock (Fig. 41.11b) and the figure of six (Fig. 41.11c). Usually at least two of these methods are used together.

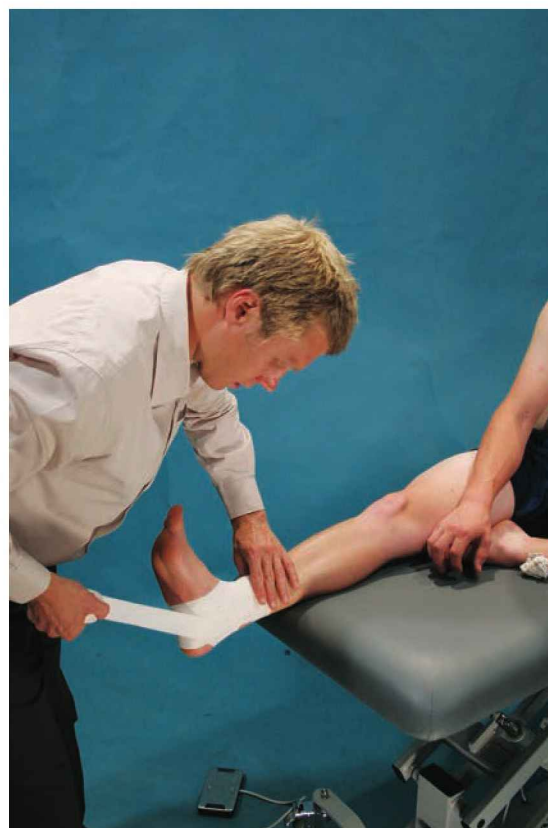
Braces have the advantage of ease of fitting and adjustment, lack of skin irritation and reduced cost compared with taping for a lengthy period. There are a number of different ankle braces available. Lace up and/or Velcro braces are the most common (Fig. 41.12).

External measures (e.g. braces/tape) act primarily by temporarily enhancing proprioception after an ankle sprain, rather than by restricting ankle range of motion.^{17–20} This support is immediately available. Nonetheless, external measures only support the impaired ankle and do not rehabilitate the underlying impaired neuromuscular function. As such, their preventive effects are limited to the period when the athlete is wearing the external measure. This means that an athlete should brace or tape for the entire 6–12 month healing period during which risk is increased to fully benefit from any preventive effects.²¹

In contrast neuromuscular training targets the underlying impairment by re-establishing and strengthening the ligament, muscles and protective reflexes of the ankle.^{22, 23} After completion of the training the athlete's increased recurrence risk is reduced and, in theory, no



(a)



(b)

Figure 41.11 Application of ankle tape (a) stirrups. After preparation of the skin, anchors are applied circumferentially. The ankle should be in the neutral position. Stirrups are applied from medial to lateral, and repeated several times until functional stability is achieved (b) heel lock. To limit inversion, taping is commenced at the front of the ankle and then angled inferiorly across the medial longitudinal arch, then diagonally and posteriorly across the lateral aspect of the heel, and then continued medially over the back of the Achilles tendon to loop back anteriorly. Tape direction is thereafter reversed to restrict eversion

further preventive means are necessary. However, these exercises do not reduce injury risk immediately with the first training session.

PRACTICE PEARL

It takes 8–10 weeks for more intensive neuromuscular training programs to achieve an effect.^{9, 24}

When external measures are employed during a period of intensive neuromuscular training after return to sport, the patient benefits from an immediate risk-reducing effect while targeting the underlying causes of an increased recurrence risk. This results in full reduction of the risk of a recurrent sprain within 10 weeks after return to sport.

Treatment of grade III injuries

Systematic reviews support non-surgical treatment of grade III ankle sprains.^{25, 26} Functional recovery (as measured by return to work) was quicker in those treated with rehabilitation, subsequent rate of ankle sprains was no different between groups, and there was more ankle stiffness in those treated surgically.²⁶

Finnish researchers compared surgical treatment (primary repair plus early controlled mobilisation) with early controlled mobilisation alone in a prospective study of 60 patients with grade III lateral ankle ligament injuries.²⁷ Of the patients treated with rehabilitation alone, 87% had excellent or good outcomes compared with 60% of patients treated surgically. Thus, early mobilisation alone provided a better outcome than surgery plus mobilisation in patients with complete tears of the lateral



(c)

Figure 41.11 cont. Application of ankle tape (c) figure of six. This is applied over stirrups. Tape runs longitudinally along the medial ankle, under the heel and is pulled up to loop back around the medial ankle as shown

ankle ligaments. Dutch investigators have reported better long-term outcomes after surgery for lateral ligament rupture compared with rehabilitation, but this conclusion was controversial.²⁸ Differences in rehabilitation protocols may explain such contradictory study results.

PRACTICE PEARL

Thus, grade III ankle injuries warrant a trial of initial conservative management over at least a 6–12 week period, irrespective of the level of the athlete.

If, despite appropriate rehabilitation and protection, there is recurrent instability or persistent pain, then surgical reconstruction of the lateral ligament is indicated.

The preferred surgical method is anatomical reconstruction using the damaged ligaments; this method produces good functional results in several studies, with



Figure 41.12 Ankle braces—a variety of braces are available

low risk of complications. The ligaments are shortened and reinserted to bone. If the ligament tissue is extremely damaged or even absent, other methods can be considered. Postoperatively, early range of motion training, such as using an air-cast stirrup, is recommended.

Following surgery, it is important for the patient to undertake a comprehensive rehabilitation program to restore full joint range of motion, strength and proprioception. The principles of rehabilitation outlined earlier are appropriate.

Anatomical reconstruction produces good clinical results in more than 90% of patients. There is, however, increased risk of inferior results in cases of very longstanding ligament insufficiency and generalised joint laxity.

LESS COMMON CAUSES

Medial (deltoid) ligament injuries

As the medial ligament is stronger than the lateral ligament, and probably because eversion is a less common mechanism of ankle sprain, medial ankle ligament injuries are less common than lateral ligament injuries.

The classical injury mechanism of a medial ligament lesion is forced eversion of the foot, such as a direct

trauma to the lateral aspect of the foot and ankle. It is relatively uncommon that an inversion trauma causes combined lateral and deltoid ligament injuries.

Physical examination reveals swelling and pain on palpation of the medial ligament with signs of talar subluxation in complete lesions. Standard radiographs are frequently normal. The sensitivity of MRI to detect medial ligament lesions is high. However, there is a risk of over-diagnosing medial ligament lesions—which can be seen after a classical inversion trauma (with lateral ligaments injuries) in 60% of the cases, but this frequently does not match the clinical findings.

Medial ligament injuries may occur together with fractures (e.g. medial malleolus, talar dome, articular surfaces) and the treatment will depend on these associated injuries. Isolated medial ligament sprains should be treated in the same manner as lateral ligament sprains. In cases of gross laxity, elite athletes can be advised to consider a surgical option.

PRACTICE PEARL

After a medial ligament injury, return to activity takes about twice as long (or more) as would be predicted were the injury on the lateral side.

Significant ankle fractures

A fracture affecting one or more of the malleoli (lateral, medial, posterior) can be difficult to distinguish from a moderate-to-severe ligament sprain because both conditions result from similar mechanisms of injury and cause severe pain and inability to weight-bear. Careful and gentle palpation can generally localise the greatest site of tenderness to either the malleoli (fracture) or just distal to the ligament attachment (sprain). Radiographs should be ordered; the Ottawa ankle rules (Fig. 41.4) are useful to avoid missing fractures or performing unnecessary investigations in these cases.

Treatment of ankle fractures (sometimes called Pott's fractures) requires restoration of the anatomy of the superior surface of the talus and the ankle mortise (inferior margins of the tibia and fibula). If the anatomy has been disrupted with dislocation of the joint surface, internal fixation is almost always required. When using stable internal fixation, early range of motion training can be started early.

Isolated undisplaced spiral fractures of the lateral malleolus (without medial ligament instability) and posterior malleolar fractures involving less than 25% of the articular surface are usually stable. These fractures can be treated symptomatically with early mobilisation using crutches in the early stages for pain relief.

Lateral malleolar fractures associated with medial instability, undisplaced medial malleolar fractures or larger undisplaced posterior malleolar fractures are potentially unstable, but may be treated conservatively with six weeks of immobilisation using a walking boot (Fig. 41.13) that extends to include the metatarsal heads. In cases of undisplaced or minimally displaced fractures, the immobilisation time may be shortened considerably, using an ankle brace and early range of motion training.

Displaced medial malleolar, large posterior malleolar, bimalleolar or trimalleolar fractures, or any displaced fracture that involves the ankle mortise, require orthopaedic referral for open reduction and internal fixation. A comprehensive rehabilitation program should be undertaken following surgical fixation. The aims of the rehabilitation program are to restore full range of motion, strengthen the surrounding muscles and improve proprioception. Refer to guidelines provided earlier.

Lateral malleolar fracture with syndesmotic injury (Maisonneuve fracture)

This injury is more common in patients presenting to emergency departments than in the sports setting, but occasionally high-impact sports injuries can cause this variant of the syndesmosis sprain.

PRACTICE PEARL

The injury involves complete rupture of the medial ligament, the AITFL (see below) and interosseous membrane, along with a proximal fibular fracture.

Surprisingly, non weight-bearing X-rays may not demonstrate the fracture, and the unstable ankle can reduce spontaneously. Urgent referral to an orthopaedic surgeon is necessary.²⁹

PERSISTENT PAIN AFTER ANKLE SPRAIN—‘THE DIFFICULT ANKLE’

Most cases of ankle ligament sprain resolve satisfactorily with treatment—pain and swelling settle and function improves. However, as ankle sprain is such a common condition, a substantial number of patients do not progress well and complain of pain, recurrent instability, swelling and impaired function 3–6 weeks after injury, or even longer. This is a common presentation in sports medicine practice and the key to successful management is accurate diagnosis. The ankle may continue to cause problems because of an undiagnosed fracture or other bony abnormality (Table 41.2). Alternatively, there may be ligament, tendon, synovial or neurological dysfunction (Table 41.3). Here we discuss a clinical approach to managing patients with the difficult ankle before detailing management of specific conditions.

Table 41.2 Fractures and impingements that may cause the ‘difficult ankle’—persistent ankle pain after ankle injury

Fractures	Bony impingements ^(a)
Anterior process calcaneus	Anterior impingement
Lateral process talus	Posterior impingement
Posterior process talus (also os trigonum fracture)	Anterolateral impingement
Osteochondral lesion	
Tibial plafond chondral lesion	
Fracture of base of fifth metatarsal	

(a) Although impingements are included here in the bony causes, pain commonly arises from soft tissue impingement between bony prominences.

Clinical approach to the difficult ankle

The clinician should take a detailed history: clarify whether pain or impairment started following an ankle sprain (the true ‘difficult ankle’) or if it is longstanding and arose without a history of injury (see Chapter 42). The patient who has had inadequate rehabilitation will usually complain of persistent pain, reduced range of motion and limitation of function with increasing activity. The clinician should determine whether the rehabilitation was adequate by asking the patient to demonstrate the exercises he or she performed in rehabilitation. Did therapy include range of motion exercises (particularly dorsiflexion), strengthening exercises (with the foot fully plantarflexed to engage the peroneal tendons/muscles) and proprioceptive retraining?³⁰

Examination of the inadequately rehabilitated ankle reveals decreased range of motion in the ankle joint (especially dorsiflexion), weak peroneal muscles and impaired proprioception. These findings can be reversed with mobilisation of the ankle joint (Fig. 41.7a), muscle strengthening (Fig. 41.8) and training of proprioception (Fig. 41.10). Other abnormalities can also cause this constellation of examination findings—remember that the ankle may be inadequately rehabilitated because of the pain of an osteochondral lesion of the talus.

If rehabilitation has been appropriate and symptoms persist, it is necessary to consider the presence of other abnormalities. Was it a high-energy injury that may have caused a fracture? Symptoms of intra-articular abnormalities include clicking, locking and joint swelling. The clinician should palpate all the sites of potential fracture very carefully to exclude that condition. See also the section on Ottawa ankle rules (Fig. 41.4).

Soft tissue injuries that can cause persistent ankle pain after sprain include chronic ligament instability, complex regional pain syndrome type 1 and, rarely, tendon dislocation or subluxation, or even tendon rupture (partial or total). Inflammation of the sinus tarsi (sinus tarsi syndrome) can also cause persistent ankle pain, but this can also occur secondary to associated fractures. Thus, even if there are features of the sinus tarsi syndrome, the clinician should still seek other injuries too.

Appropriate investigation is a key part of management of patients with the difficult ankle. MRI is preferred to bone scan as it can detect bony and soft tissue abnormalities. Clinicians must remember that a subluxing tendon can appear normal on MRI.

PRACTICE PEARL

Isotope activity in the bone phase is normal in soft tissue problems and increased if bony damage is present.

Osteochondral lesions of the talar dome

An osteochondral lesion of the talus is an important cause of residual pain after an ankle sprain. It is defined as the separation of a fragment of articular cartilage, with or without subchondral bone. Its incidence after an ankle sprain has been reported to be around 6.5%, and is probably underestimated because these lesions often remain undetected. The grading of osteochondral fractures of the talar dome is shown in Figure 41.15.

History

Osteochondral fractures of the talar dome usually occur in association with ankle sprains when there is a compressive

Table 41.3 Ligamentous, tendon and neurological causes of the ‘difficult ankle’—persistent ankle pain after ankle injury

Atypical sprains	Tendon injuries	Other soft tissue and neural abnormalities
Chronic ligamentous instability	Chronic peroneal tendon weakness	Inadequate rehabilitation
Medial ligament sprain	Peroneal tendon subluxation/rupture	Chronic synovitis
Syndesmosis sprain (AITFL sprain)	Tibialis posterior tendon subluxation/rupture	Sinus tarsi syndrome
Subtalar joint sprain		Complex regional pain syndrome type 1



Figure 41.13 A CAM (Controlled Ankle Movement) walker boot is a useful adjunct to rehabilitation for several different ankle injuries

component to the inversion injury, such as when landing from a jump. The talar dome is compressed by the tibial plafond, causing damage to the osteochondral surface. The lesions are usually in the superomedial corner of the talar dome, and are less common superolaterally.

Large fractures may be recognised at the time of injury. The fracture site will be tender and may be evident on plain radiography (Fig. 41.14a).

Often an osteochondral lesion of the talus is not detected initially and the patient presents later with unremitting ankle aching and locking or catching, despite appropriate treatment for an ankle sprain. The patient often gives a history of initially progressing well following a sprain, but then developing symptoms of increasing pain and swelling, stiffness and perhaps catching or locking as activity is increased. Reduced range of motion is often a prominent symptom. The reason why an osteochondral lesion of the talus produces these symptoms is poorly understood. It has been hypothesised that a focal loss of structural support of the talar dome or cyclical change in intra-osseous pressure with gait, triggers pain receptors in the richly innervated subchondral bone.

Examination

Examination with the patient's foot plantarflexed at 35° to rotate the talus out of the ankle mortise may reveal

tenderness of the dome of the talus. This may, however, be difficult due to pain. The diagnosis of osteochondral lesion of the talus requires a high index of suspicion. Focal chondral and osteochondral defects may eventually progress to ankle joint osteoarthritis.³¹

Investigations

Plain radiographs often appear normal or show only minor changes resulting in a delayed diagnosis. If osteochondral lesion of the talus is suspected, an MRI (Fig. 41.14b) should be requested.

A CT scan (Fig. 41.14c) can be used to determine the exact degree of skeletal injury.

MRI provides anatomical and pathological data, and is the investigation of choice. MRI has a benefit of being positive in grade I lesions. The grading of osteochondral fractures of the talar dome is shown in Figure 41.15.

Treatment

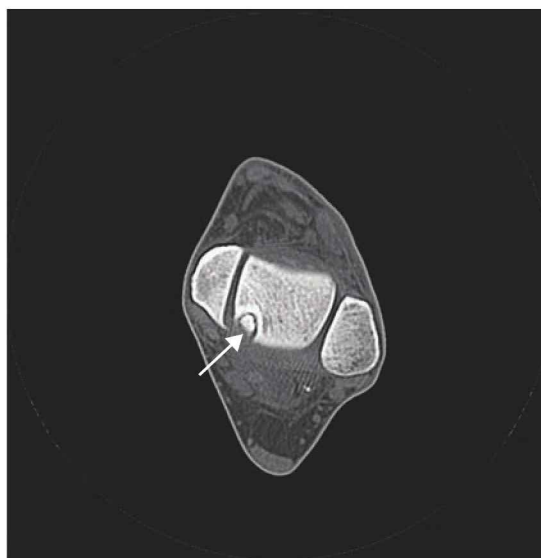
The treatment of an osteochondral lesion of the talus is complex and numerous non-surgical and surgical options including marrow stimulation techniques and tissue transplantation have been reported. All options have some limitations, mainly due to the poor regenerative capacity of the articular cartilage and the limited access to the ankle joint.



(a)



(b)



(c)

Figure 41.14 Osteochondral lesion of the talar dome
(a) radiograph (arrow) (b) MRI (circle) (c) CT scan (grade IV)

Chronic grade I and II lesions should be treated conservatively. Patients should avoid activities that cause pain and ride an exercise bike with low resistance. Cast immobilisation has now been replaced with early joint motion without significant loading to promote articular cartilage. If there is pain, symptoms of clicking, locking or giving way persisting after 2–3 months of conservative

management, ankle arthroscopy is indicated in order to remove or fix loose bodies. A grade IIa, III or IV lesion also requires arthroscopic removal of the separated fragment or cyst, and curetting and drilling of the fracture bed down to bleeding bone. After surgical treatment of osteochondral lesions, a comprehensive rehabilitation program is required. Tibial plafond chondral lesions (see below) are managed identically.

Clinical outcome studies have identified some prognostic parameters, such as the size of the lesion. However, there is only limited information available to predict the individual patient outcomes when treating osteochondral lesions of the talus.

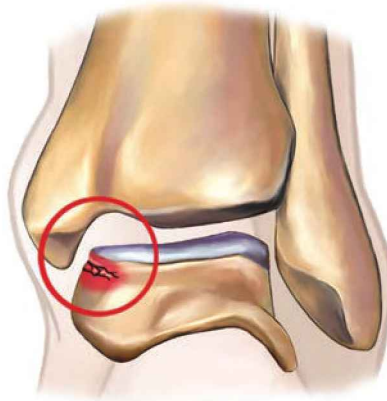
Avulsion fracture of the base of the fifth metatarsal

Inversion injury may result in an avulsion fracture of the base of the fifth metatarsal. This benign fracture can occur in isolation or, more commonly, in association with a lateral ligament sprain. The fracture results from traction of the peroneus brevis tendon attachment at the base of the fifth metatarsal.

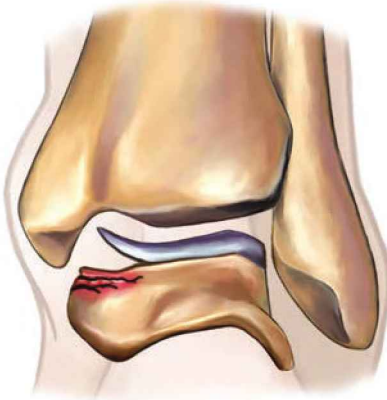
Plain radiographs should be examined closely. Avulsion fracture is characterised by its involvement of the joint surface of the base of the fifth metatarsal (Fig. 41.16).

A potentially confusing fracture is the fracture of the proximal diaphysis of the fifth metatarsal that does not involve any joint surfaces. This fracture is known as the Jones fracture and may require internal fixation (Chapter 43).

Grade I Subchondral fracture Investigation: MRI



Grade II Chondral fracture Investigation: CT/MRI



Grade IIa Subchondral cyst Investigation: CT/MRI



Grade III Chondral fracture with separated but not displaced fragments Investigation: CT/MRI



Grade IV Chondral fracture with separated and displaced fragment(s) Investigation: X-ray/CT/MRI



Figure 41.15 Grading of osteochondral fracture of the talar dome

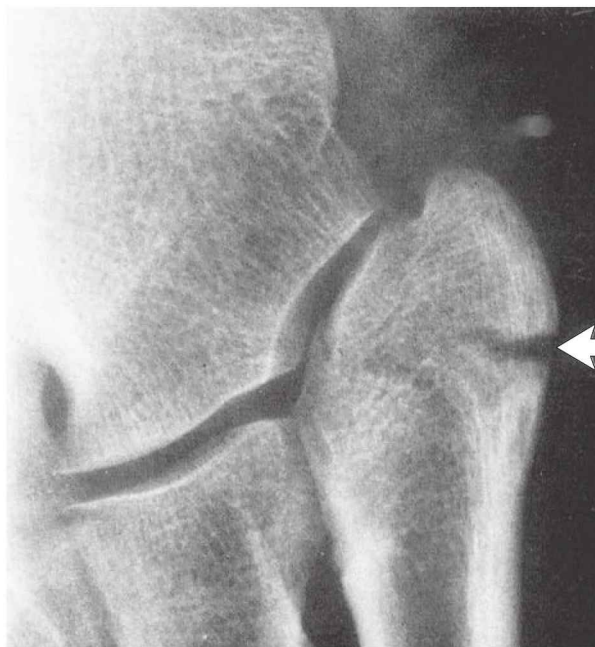


Figure 41.16 Avulsion fracture of the base of the fifth metatarsal

PRACTICE PEARL

It is important to distinguish between the benign fracture of the base of the fifth metatarsal and a Jones fracture. Refer to Chapter 43 for more information.

Although the mechanism can appear to be one of 'acute' injury, in most cases the Jones fracture is a result of repetitive overuse (i.e. a stress fracture) of the proximal diaphysis of the fifth metatarsal.

Other fractures

A number of other fractures can occur as a result of acute ankle injuries (Fig. 41.17), alone or in association with ligamentous injury. They may be undetectable or subtle on plain radiograph.

Fractured lateral talar process

The lateral talar process is a prominence of the lateral talar body with an articular surface dorsolaterally for the fibula and inferomedially for the anterior portion of the posterior calcaneal facet (Fig. 41.17). Patients with a fracture of this process may present with ankle pain, swelling and inability to weight-bear for long periods. Examination

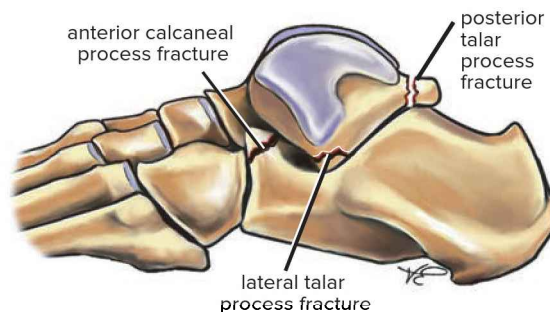


Figure 41.17 Fractures around the talus and calcaneus

reveals swelling and bruising over the lateral aspect of the ankle and tenderness over the lateral process, immediately anterior and inferior to the tip of the lateral malleolus. The fracture is best seen on the mortise view radiograph of the ankle. Undisplaced fractures may be treated in a short leg cast. Fractures displaced more than 2 mm require either primary excision or reduction and internal fixation. Comminuted fractures may require primary excision, which is a safe treatment, followed by short immobilisation only.

Fractured anterior calcaneal process

Fractures of the anterior calcaneal process may cause persistent pain after an ankle sprain. Palpation of the anterior calcaneal process, just anterior to the opening of the sinus tarsi (Fig. 41.17), is painless in patients with a tear of the ATFL, but will cause considerable pain in those with a fracture of the anterior process. If plain radiographs (including oblique foot views) fail to show a fracture that is suspected clinically, MRI or CT is indicated (Fig. 41.18).

If the fracture is small, symptomatic treatment may suffice. If large, it requires four weeks of non weight-bearing cast immobilisation or surgical excision of the fragment.

Tibial plafond chondral lesions

Tibial plafond (the inferior tibial articular surface) injuries may occur with vertical compression forces, such as a fall from a height. However, they can also result from straightforward ankle sprains. The patient complains of difficulty weight-bearing, and examination reveals swelling and restricted dorsiflexion. As with talar dome lesions, plain radiograph can be normal, so MRI or CT are necessary to demonstrate the lesion (Fig. 41.19).

If imaging and clinical features are consistent with bony damage, arthroscopic debridement, including microfracturing is indicated but ankle pain can persist for months to a year, even after treatment.



Figure 41.18 Sagittal CT image showing an anterior process calcaneal fracture

Fractured posterior process of the talus

Posterior process fractures occur usually as a result of forced plantar flexion injuries and are even less common than lateral process fractures (Fig. 41.17). This mechanism is often seen during kicking and has been reported in fencing.

Routine AP and lateral radiographs may not show the acute fracture and can be wrongly interpreted. CT scanning remains the mainstay of diagnosis, but a high index of suspicion is required.

Most of these injuries will be treated initially with conservative treatment. Acute fractures of the posterior process of the talus may require surgical excision.

Medial tubercle fractures

Fractures of the medial tubercle of the posterior process talus are rare in sports. They can occur due to a dorsiflexion avulsion by the posterior talotibial ligament (posterior aspect of the deltoid ligament) and eversion (Cedell fracture), or by direct traumatic compression.

In contrast to lateral tubercle injuries, pain and swelling is usually present between the Achilles tendon and the medial malleolus, but there may be limited pain on walking or moving the ankle. It is difficult to visualise fractures of the medial tubercle on plain AP and lateral radiographs, and it has been suggested that the addition of two oblique views at 45° and 70° of external rotation may



Figure 41.19 Sagittal T2 MRI image of an anterior tibial plafond cartilage lesion with a loose fragment of cartilage in the joint

significantly aid detection prior to resorting to a CT or MR imaging. The fracture can be treated arthroscopically with a posterior approach.

Impingement syndromes

Ankle impingement syndromes are usually the result of overuse, but can occur following acute ankle injuries. For example, ballet dancers often suffer posterior impingement following lateral ankle sprain, due to the presence of an os trigonum, processus posterior of the talus and/or posterior osteophytes. In the acute ankle a blocked kicking action can cause posterior impingement. Posterior impingement syndrome is discussed in detail in Chapter 40. Anterior and anterolateral impingement syndromes and their treatment are discussed in Chapter 42.

Tendon dislocation or rupture

Dislocation or longitudinal rupture of the peroneal tendons (in most cases peroneus brevis tendon) can cause persistent lateral ankle symptoms. Tibialis posterior injury can cause similar symptoms on the medial side but is very uncommon in sport.

Dislocation of the peroneal tendons

The peroneal tendons are situated behind the lateral malleolus and held there by the superior peroneal

retinaculum. They are occasionally dislocated as a result of forceful passive dorsiflexion with rupture of the retinaculum from the posterior edge of the lateral malleolus. This may occur when a skier catches a tip and falls forward over the ski. Rupture of the peroneal retinaculum allows one or both of the tendons to move out of their groove. The dislocated tendon(s) may remain in a dislocated position or spontaneously relocate and subsequently become prone to recurrent subluxation. Examination reveals tender peroneal tendons that can be dislocated by the examiner, especially with ankle plantarflexion.

Treatment of dislocation of peroneal tendons is surgical replacement of the tendons in the peroneal groove and repair of the retinaculum, using bone anchors or drill holes. If the peroneal groove is shallow, then retinacular repair should be accompanied by deepening of the groove or rotation of the malleolus (not often recommended). Soft tissue repair, however, produces a good result in most cases.

Dislocation of the tibialis posterior tendon

Dislocation of the tibialis posterior tendon is extremely rare in sport. It occurs with ankle dorsiflexion and inversion, where forceful contraction of the tibialis posterior muscle pulls the tendon out of its retinaculum using the malleolus as a fulcrum. The patient may complain of moderate medial ankle pain and inability to weight-bear. Examination reveals swelling and bruising around the medial malleolus with tenderness along the course of the tibialis posterior tendon. The tendon can be subluxed anteriorly and subsequently relocated posteriorly with the foot in the fully plantarflexed position. The diagnosis is clinical, but ultrasonography or MRI (Fig. 41.20) may reveal fluid around the tendon.

Immediate surgical treatment is indicated to minimise the time that the tendon is dislocated while permitting primary repair of the flexor retinaculum and reattachment of the tibialis posterior sheath.³² Postoperatively the ankle is immobilised in a CAM walker for a total of six weeks' non weight-bearing. Once the walking boot is no longer required an ankle brace can be used to support the ankle with active ankle motion permitted three times daily while taking care to avoid resisted inversion. Weight-bearing can recommence at six weeks under physiotherapy supervision followed by strengthening and functional rehabilitation.

Rupture of the tibialis posterior tendon

An athlete with a ruptured tibialis posterior tendon presents with pain in the region of the tubercle of the navicular extending to the posterosuperior border of the medial malleolus and along the posteromedial tibial border. Examination reveals thickening or absence (less

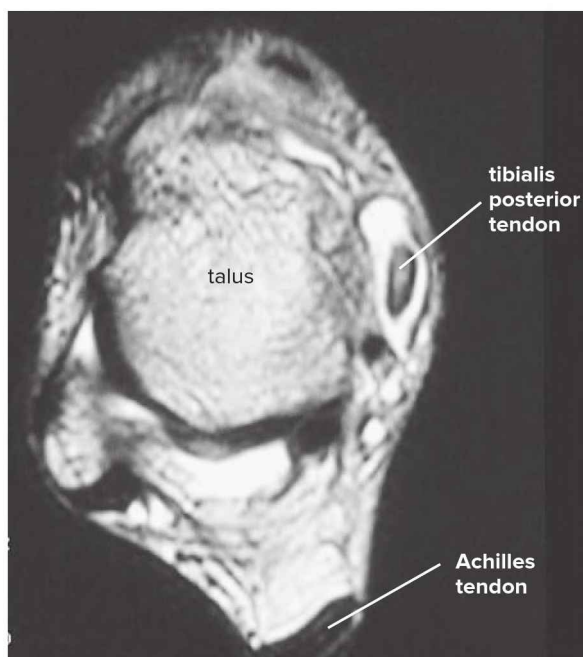


Figure 41.20 MRI appearance (T2-weighted) shortly after tibialis posterior tendon dislocation shows the tibialis posterior tendon (dark) in cross-section surrounded by abnormal fluid (high signal intensity). The tendon is in its normal position during this examination. If imaging had been delayed sufficiently, fluid would have been absent and the MRI appearance may have been normal

frequent) of the tibialis posterior tendon and inability to raise the heel. A flattened medial arch is a classic sign. MRI is the investigation of choice in this condition, although ultrasound may also be helpful. However, there is low correlation between MRI findings and symptoms. Surgical repair, often including major reconstruction of the midfoot anatomy is indicated as the tibialis posterior tendon is essential to maintain the normal medial arch of the foot.

Other causes of the difficult ankle

Anteroinferior tibiofibular ligament injury/syndesmotic injury

The distal tibiofibular joint is a fibrous syndesmotic articulation consisting of the concave surface of the distal tibia and convex shape of the distal fibula.³³ The syndesmosis is a complex ligamentous structure including three different portions. The anterior syndesmosis consists of the anterior inferior tibiofibular ligament (AITFL); the middle part consists of the interosseous ligament (IOL); and the posterior syndesmosis includes the posterior inferior tibiofibular and the transverse ligaments (PITFL and TL). Each structure plays a critical role to provide

stability of the ankle mortise. The resistance to diastasis comes from the PITFL and TL for 40–45%, the AITFL for 35% and the IOL for 20–25%.³⁴

Injuries to the distal tibiofibular syndesmosis (Fig. 41.21) account for 11–17% of the ankle sprains in athletic populations.³⁵ Acute syndesmotic injuries have been reported especially in football and skiing. They can range from stable minor sprains to significant fractures (e.g. Maisonneuve fracture) of the distal fibula with combined syndesmotic disruption. Syndesmotic injuries usually present with more pain and disability than a typical lateral ankle sprain.

The mechanism of syndesmotic injury can be an isolated hyperdorsiflexion force, but is most commonly due to an external rotation injury, in combination with axial loading of the ankle. In patients with a combined deltoid injury or medial malleolar fracture, an abduction force is the main cause of the associated syndesmotic lesion.

Syndesmotic injuries can be isolated or associated with bony or ligamentous ankle injuries. The term 'isolated' is commonly used for syndesmotic injuries without ankle fractures, although concomitant ligamentous or soft tissue injuries and tibiofibular avulsion fractures may be present.³⁶

If there is pain on palpation over the anterior and posterior tibiofibular ligaments, the more proximal the pain, the greater the extent of the lesion. It is also important to check the entire fibula up to the knee if there is any suspicion of syndesmotic/Maisonneuve

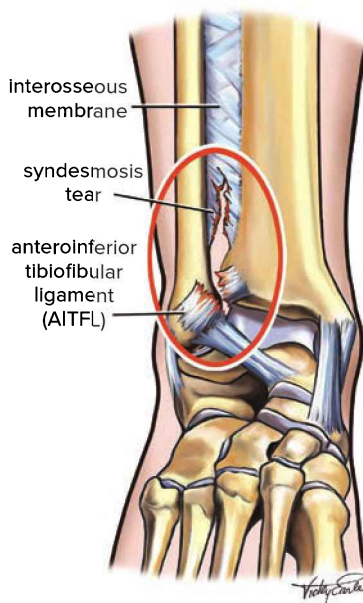


Figure 41.21 Anatomy of a syndesmosis sprain. This injury may be associated with medial malleolar fracture (not illustrated)

fracture. Swelling is less marked in syndesmotic injury compared with a normal ankle sprain and bruising is usually more proximal to the ankle joint.

PRACTICE PEARL

There is no pathognomonic clinical test to evaluate a syndesmotic injury but the external rotation test is the most helpful test. The test is performed by stabilising the leg with the knee flexed at 90° and then rotating the foot externally.

A positive test will elicit pain at the syndesmosis. During physical examination, it is important to palpate the medial ankle in addition to determine the integrity of the deltoid ligament, since it can highlight a more significant unstable syndesmotic injury.

Radiographic imaging must include an AP view and a mortise view of the syndesmosis to rule out the tibiofibular clear space, medial clear space overlap, tibial width and fibular width. In acute syndesmotic lesions, MRIs have a sensitivity and specificity that are about 100%.³⁷ Furthermore, injury to the distal tibiofibular syndesmosis has a significant association with a number of secondary findings on MRI, including bone bruises, osteochondral lesions, tibiofibular joint congruity and height of the tibiofibular recess.

Acute ankle injuries with an isolated lesion of the syndesmosis should be classified as stable or unstable.

- The stable ankle sprain is characterised by a lesion of the AITFL with or without interosseous ligament (IOL) with a continent deltoid ligament.
- The unstable ankle sprain should be classified as latent or frank diastasis. The latent diastasis is characterised by a rupture of the AITFL with or without IOL and deltoid ligament. It can be detected on MRI and/or arthroscopic assessment. The frank diastasis is characterised by a rupture of all the syndesmotic and deltoid ligaments, and it is visible on standard radiographs.³³

Syndesmotic sprains without instability should be treated non-operatively with a CAM walker (Fig. 41.13) or brace. Syndesmosis injuries frequently require almost twice as much time rehabilitating compared with patients who have lateral ankle sprains. An acute syndesmotic ligament rupture with instability of the ankle should be managed operatively.³³

Post-traumatic synovitis

Some degree of synovitis will occur with any ankle injury due to the presence of blood within the joint. This

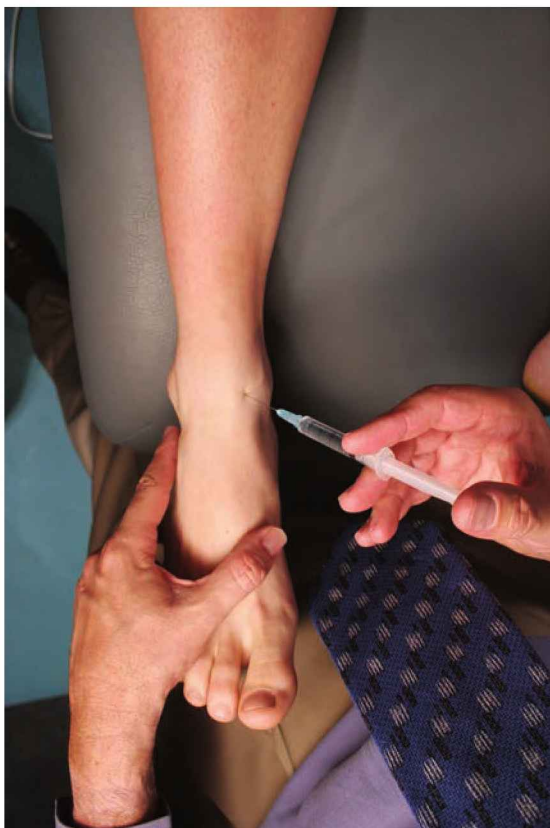


Figure 41.22 Corticosteroid injection into the ankle joint in the treatment of post-traumatic synovitis. The needle is inserted medial to the tibialis anterior tendon and directed posterolaterally

usually resolves in a few days, but may persist if there is excessive early weight-bearing, typically in athletes eager to return to training soon after their ankle sprain, or due to insufficient rehabilitation. These athletes will often develop persistent ankle pain aggravated by activity and associated with swelling. Synovitis of the ankle joint is also seen in athletes who have chronic mild instability because of excessive accessory movement of the ankle joint during activity.

Treatment of synovitis includes NSAIDs, rest from aggravating activity and local electrotherapy. A corticosteroid injection into the ankle joint (Fig. 41.22) may be required. An injection should be followed by 48 hours of limited weight-bearing and gradual resumption of activity. Sometimes arthroscopy with synovectomy may be indicated.

When synovitis is associated with a degree of chronic instability, treatment involves taping or bracing. Such patients can have significant relief when wearing a brace for



Figure 41.23 Injection into the sinus tarsi. The lateral opening of the sinus is maintained when the foot is inverted. The needle is directed towards the tip of the medial malleolus

activities of daily living as well as sport. These patients may also benefit from ankle ligament reconstructive surgery.

Sinus tarsi syndrome

Sinus tarsi syndrome may occur as an overuse injury secondary to excessive subtalar pronation (Chapter 42) or following an ankle sprain. Pain occurs at the lateral opening of the sinus tarsi (Fig. 41.1b). The pain is often more severe in the morning and improves on warming up.

Forced passive inversion and eversion may both be painful. The most appropriate aid to diagnosis is to monitor the effect of injection of a local anaesthetic agent into the sinus tarsi under fluoroscopy (Fig. 41.23).

Treatment consists of relative rest, NSAIDs, electrotherapeutic modalities, subtalar joint mobilisation and taping to correct excessive pronation if present. If conservative management is unsuccessful, injection of corticosteroid and local anaesthetic agents may help resolve the inflammation.

Complex regional pain syndrome type 1

Complex regional pain syndrome (CRPS) type 1, formerly known as RSD (Chapters 5, 6 and 7), may occasionally complicate ankle injury. Initially, it appears that the patient with a 'sprained ankle' is improving but then symptoms begin to relapse. The patient complains of increased pain, swelling recurs and the skin may become hot or, more frequently, very cold. There may also be localised sweating, discoloration and hypersensitivity.

As early treatment substantially improves the prognosis in CRPS type 1, early diagnosis is imperative. Initial radiographs are normal. Later, patchy demineralisation occurs and this can be seen as regions of decreased opacity on plain radiograph and areas of increased uptake on bone

scan or increased signal on MRI.³⁸ Tests of sympathetic function may confirm the diagnosis.³⁸

It is most important that the specific nature of the condition be explained to the patient as it may be very painful, even at rest. It remains difficult to treat and there have been few controlled treatment trials for established CRPS type 1.³⁹ Physiotherapy may play a role⁴⁰ and ultrasound and hydrotherapy may facilitate range of movement exercises. Gabapentin, an anticonvulsant with a proven analgesic effect in various neuropathic pain syndromes, has shown mild efficacy as treatment for pain in patients with CRPS

type 1.⁴¹ As CRPS type 1 is associated with regional osteoclastic overactivity (excessive bone turnover, as shown by increased uptake on radionuclide bone scan), a bisphosphonate medication (alendronate) was trialled in 39 patients.⁴² In contrast to placebo-treated patients, all of the alendronate-treated patients had substantially reduced pain and improved joint mobility as early as the fourth week of treatment.

If the pain does not settle, chemical or surgical blockade is indicated. However, a Cochrane systematic review failed to support this therapy for relieving pain.⁴³ CRPS type 1 remains a very difficult condition to treat.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

